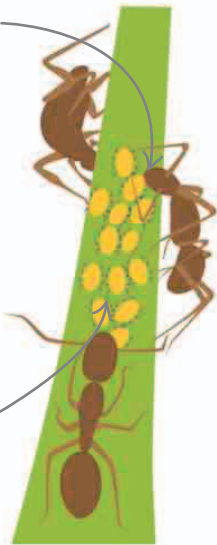


Symbiosis

When animals of two species cooperate with each other, the relationship is known as symbiosis. There are two types. In mutualistic relationships, both partners benefit from the actions of the other. Commensal relationships are rarer. They involve one animal benefiting from the association, while the other receives no benefit, but is not harmed either.

ants stroke the aphid to get a drink of honeydew

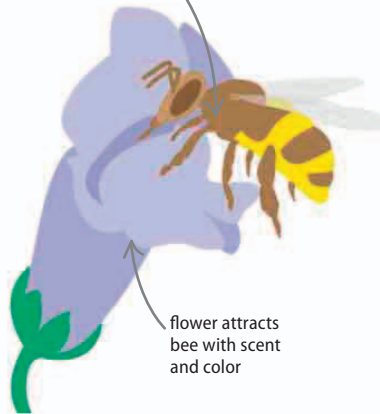
aphids suck sap from plant stem



△ Ants and aphids

Aphids produce a sweet urine called honeydew. Ants protect a herd of aphids from predators, in order to feed on the aphids' honeydew.

bee stores nectar in stomach for conversion to honey in the hive



△ Honeybee and flower

Flowers rely on honeybees to transfer their pollen to another plant. In return, the bees feed on nectar supplied by the plant.

shark gets cleaned by the pilot fish



△ Pilot fish and shark

The small fish follow a large predator and snap up the leftovers from its meals, keeping the predator clean in the process.

impala tolerates the oxpecker because the bird keeps its coat clean of parasites

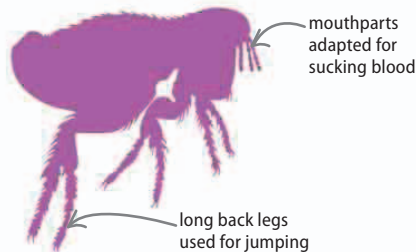


△ Oxpecker and impala

An oxpecker bird lives on the back of a large herbivore, such as an impala. The bird feeds on ticks and insects living on the larger animal's hair and skin.

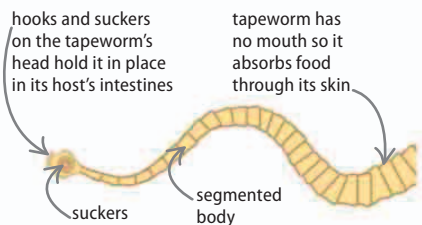
Parasites

A parasite is an organism that lives on or inside another, known as the host. The parasite either eats the body of the host or consumes some of its food. The host is disadvantaged by the relationship, but is not killed—if it was, the parasite would soon die as well in many cases. A parasitoid is an animal that does kill its host, generally as a larva eating it alive. Once the host is dead, the parasitoid takes on an independent mode of life (see page 91).



△ Flea

This insect is an ectoparasite, meaning it lives on the outside of the host. Fleas suck the blood of their hosts, moving to new ones in great leaps.



△ Tapeworm

This flatworm is an endoparasite, meaning it lives inside its host. Egg-carrying segments break off the worm and end up in the host's droppings, where they hatch and spread.